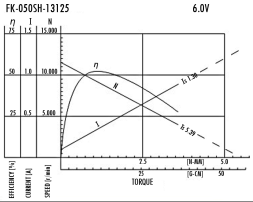
- All-metal spur gearhead mated to a Mabuchi FK-050 motor.
- 9 different gear ratios: 11.1, 22.2, 33.3, 50, 83.3 (discontinued, dwindling stock), 100, 200, 300, and 600 :1. One twice as fast, and one the Gold Sparks!
- Weight between 28 - 32grams (0.99 - 1.13oz).

These motors make it super simple to get a robot moving! Just pick up a pair of Foam Wheels and Twist Hubs to mount them, solder on and your motors are ready to be driven by radio control! For mounting the motors, use 2-56 screws and our custom motor mounts. We 3mm bearing blocks for supporting the shaft in heavier robots or weapon systems, and Flat Motor Mounts for when you want to face-mo

Note - For strenuous conditions (such as robot combat), we recommend removing the gearbox and checking that the motor vent hole dc a gearbox pin. If it does, simply rotate the gearbox 180 degrees and reattach. Use Loctite on the screws that hold the gearbox to the mo they cannot separate during competition!

Specifications:

Motor (Mabuchi FK-050SH-13125) Specifications	**Physical Specifications**
Operating Voltage: 3V - 18.5V (>10V decreases motor life) Nominal Voltage: 6V, but most commonly run at 11.1V. (14.8V runs fine but support the shaft because the gears are taking more abuse.) No Load RPM: 11530rpm No Load Current: 0.09A Stall Current: 1.3A Stall Torque: 0.0461kg-cm (0.64oz-in) Kt: 0.0352kg-cm/A (0.49oz-in/A) Kv: 1573rpm/V Peak Efficiency: 52% RPM @ Peak Eff: 9100 Current @ Peak Eff: 0.34A Click herefor the motor torque, current, rpm and efficiency curve. 	Gearbox Lengths: ratios 11: 11.5mm (0.45in) ratios 22, 33, 50: 12.9mm (0.51in) ratios 83, 100: 14.3mm (0.56in) ratios 200, 300: 16.4mm (0.65in) ratios 600: 17.9mm (0.70in) Motor Length: 28.4mm (1.12in) Motor weight (no gearbox): 17g (0.60oz) Gearbox Diameter: 16mm (0.63in) Shaft Diameter: 3mm (0.12in) with flat along length Shaft Length: 38mm (1.5in), and now also 22mm (0.87in) 50:1 ratios for Viper spares Mounting Holes (2): #2-56 spaced 11mm (0.433in) apart -do not use screws that protrude more than 2mm (0.078in Weight: - ratios 11: 28g (0.99oz) - ratios 22, 33, 50, 83, 100: 29g (1.02oz) - ratios 200, 300: 31g (1.09oz) - ratios 600: 32g (1.13oz)

All ratio gearboxes can be interchanged because they use the same 14T pinion gears on the motor.
So (for example) you can remove the 50:1 ratio gearbox from its motor and replace it with a 33:1 ratio gearbox for more speed - without desolder the motor! (Might want to swap the stickers so you know what's in the robot!)

No Load Speed @ Gear Ratio/Voltage

Speed (rpm)	Voltage						
Gear ratio	5	6	7.4	11.1	14.8	18.5	22.2
11.1 :1	709	850	1049	1573	2097	2622	3146
22.2 :1	354	425	524	786	1049	1311	1573
33.3 :1	236	283	350	524	699	874	1049
50 :1	157	189	233	349	466	582	698
83.3 :1	94	113	140	210	279	349	419
100 :1	79	94	116	175	233	291	349
200 :1	39	47	58	87	116	146	175
300 :1	26	31	39	58	78	97	116
600 :1	13	16	19	29	39	49	58

(Click to Enlarge)

Motors have been tested to work up to 22.2V (6 lipoly cells), but don't expect the motor to run that high forever!

Maximum Torque @ Ratio/Voltage (kg-cm) (oz-

Torque (kg-cm)	Voltage						
Gear ratio	5	6	7.4	11.1	14.8	18.5	22.2
11.1 :1	0.35	0.42	0.52	0.77	1.03	1.29	1.55
22.2 :1	0.70	0.84	1.03	1.55	2.07	2.58	3.10
33.3 :1	1.05	1.26	1.55	2.32	3.10	3.87	4.65
50 :1	1.57	1.89	2.33	3.49	4.65	5.82	6.98
83.3 :1	2.62	3.14	3.88	5.81	7.75	9.69	11.63
100 :1	3.14	3.77	4.65	6.98	9.30	11.63	13.96
200 :1	6.29	7.54	9.30	13.96	18.61	23.26	27.91
300 :1	9.43	11.32	13.96	20.93	27.91	34.89	41.87
600 :1	18.86	22.63	27.91	41.87	55.82	69.78	83.74

(Click to Enlarge)

Exceeding 9.72kg-cm (135oz-in) will damage the output gear stage. Do not stall the motors indicated voltage.

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